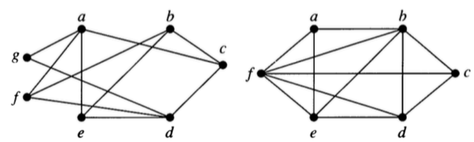
6 Bipartite Graphs

6.1 Definition

A simple graph G is **biparatite** if its vertices can be partitioned into two disjoint subsets V_1 and V_2 and every edge connects a vertex in V_1 with a vertex in V_2. The pair $\langle V_1, V_2 \rangle$ is called a **bipartition** of G.

6.2 Examples

1. Which of the following are bipartite?



G H

2. For what values of n are the following bipartite?

   a. K_n
   b. C_n
   c. Q_n

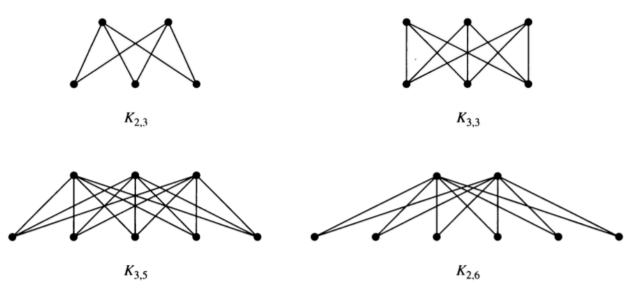
3. Can a bipartite graph have more than one bipartition? If so, give an example. If not, explain why not.

4. Describe an algorithm for checking whether a graph is bipartite. How efficient is your algorithm?

6.3 Complete Bipartite Graphs

A complete biparite graph $K_{m,n}$ has two disjoint sets of vertices V_1 and V_2 with m and n elements. There is an edge between x and y if and only if $x \in V_1$ and $y \in V_2$ or $x \in V_2$ and $y \in V_1$.

Here are some examples.



5. Draw the following graphs: $K_{2,2}$, $K_{4,2}$, $K_{4,3}$